

Matrix Multiplication

Exercise 1

$$\begin{array}{cc} & B \\ & \begin{array}{|c|c|} \hline 5 & 2 \\ \hline \end{array} \\ A \begin{array}{|c|} \hline 3 \\ \hline \end{array} & \begin{array}{|c|c|} \hline 15 & \text{ } \\ \hline \end{array} \\ & A \cdot B \end{array}$$

Matrix Multiplication

Exercise 2

$$\begin{array}{c} A \end{array} \begin{array}{|c|} \hline 3 \\ \hline 2 \\ \hline \end{array} \begin{array}{c} B \\ \begin{array}{|c|} \hline 4 \\ \hline \end{array} \end{array} \begin{array}{|c|} \hline 12 \\ \hline \end{array}$$

$A \cdot B$

8

Matrix Multiplication

Exercise 3

$$\begin{matrix} & & B \\ & & \begin{array}{|c|c|} \hline 1 & -1 \\ \hline \end{array} \\ A & \begin{array}{|c|} \hline 1 \\ \hline -1 \\ \hline \end{array} & \begin{array}{|c|c|} \hline 1 & -1 \\ \hline -1 & \end{array} \\ & & A \cdot B \end{matrix}$$

Matrix Multiplication

Exercise 4

$$\begin{matrix} & & B \\ & & \begin{bmatrix} 2 & -2 \end{bmatrix} \\ A & \begin{bmatrix} 3 \\ 2 \end{bmatrix} & \begin{bmatrix} 6 & -6 \\ 4 & \end{bmatrix} \\ & & A \cdot B \end{matrix}$$

-4

Matrix Multiplication

Exercise 5

$$\begin{matrix} & & B \\ & & \begin{bmatrix} 3 & -5 \end{bmatrix} \\ A & \begin{bmatrix} 1 \\ \end{bmatrix} & \begin{bmatrix} 3 & -5 \\ 0 & 0 \end{bmatrix} \\ & & A \cdot B \end{matrix}$$

0

Matrix Multiplication

Exercise 6

$$\begin{matrix} & & B \\ & & \begin{bmatrix} 4 & -2 \end{bmatrix} \\ A & \begin{bmatrix} 1 \\ \end{bmatrix} & \begin{bmatrix} 4 & -2 \\ -4 & 2 \end{bmatrix} \\ & & A \cdot B \end{matrix}$$

-1

Matrix Multiplication

Exercise 7

$$\begin{matrix} & & B \\ & & \begin{array}{|c|c|} \hline 1 & 3 \\ \hline 3 & 2 \\ \hline \end{array} \\ \\ A & \begin{array}{|c|c|} \hline 1 & 1 \\ \hline 1 & -1 \\ \hline \end{array} & \begin{array}{|c|c|} \hline 4 & \text{ } \\ \hline -2 & 1 \\ \hline \end{array} \\ & & A \cdot B \end{matrix}$$

Matrix Multiplication

Exercise 8

$$\begin{matrix} & & & B \\ & & \begin{array}{|c|c|} \hline 2 & 5 \\ \hline 4 & 3 \\ \hline \end{array} \\ & A & \begin{array}{|c|c|} \hline 1 & 1 \\ \hline 1 & -1 \\ \hline \end{array} \end{matrix} \quad \begin{array}{|c|c|} \hline 6 & 8 \\ \hline \text{ } & 2 \\ \hline \end{array}$$

$A \cdot B$

-2

Matrix Multiplication

Exercise 9

$$\begin{matrix} & & & B \\ & & \begin{array}{|c|c|} \hline 1 & 1 \\ \hline -1 & 1 \\ \hline \end{array} \\ & A & \begin{array}{|c|c|} \hline 3 & 2 \\ \hline 4 & 5 \\ \hline \end{array} & \begin{array}{|c|c|} \hline 1 & 5 \\ \hline \text{ } & 9 \\ \hline \end{array} \\ & & A \cdot B \end{matrix}$$

-1

Matrix Multiplication

Exercise 10

$$\begin{matrix} & & B \\ & & \begin{array}{|c|c|} \hline 1 & 1 \\ \hline -1 & 1 \\ \hline \end{array} \\ \\ A & \begin{array}{|c|c|} \hline 5 & 5 \\ \hline -2 & 2 \\ \hline \end{array} & \begin{array}{|c|c|} \hline 0 & 10 \\ \hline -4 & \text{ } \\ \hline \end{array} \\ & & A \cdot B \end{matrix}$$

0

Matrix Multiplication

Exercise 11

B

1	1	-1	-1
-1	1	1	-1

A

3	1
2	4

2	4		-4
-2	6	2	-6

$A \cdot B$

-2

Matrix Multiplication

Exercise 12

$$\begin{matrix} & & & & B \\ & & & \begin{array}{|c|c|c|c|} \hline 4 & 3 & 2 & 1 \\ \hline 1 & 2 & 1 & 2 \\ \hline \end{array} \\ & & & & \\ A & \begin{array}{|c|c|} \hline 1 & 1 \\ \hline 1 & -1 \\ \hline 1 & 0 \\ \hline \end{array} & \begin{array}{|c|c|c|c|} \hline 5 & 5 & 3 & 3 \\ \hline 3 & 1 & & -1 \\ \hline 4 & 3 & 2 & 1 \\ \hline \end{array} \\ & & A \cdot B \end{matrix}$$

Matrix Multiplication

Exercise 13

$$\begin{matrix} & & & B \\ & & \begin{array}{|c|c|} \hline 2 & 1 \\ \hline 3 & 5 \\ \hline 4 & -1 \\ \hline \end{array} \\ & A & \begin{array}{|c|c|c|} \hline 1 & 0 & 1 \\ \hline 0 & 1 & 1 \\ \hline \end{array} \begin{array}{|c|c|} \hline 6 & 0 \\ \hline 7 & \text{ } \\ \hline \end{array} \\ & & A \cdot B \end{matrix}$$

Matrix Multiplication

Exercise 14

$$\begin{matrix} & & & B \\ & & & \begin{array}{|c|c|} \hline 1 & 1 \\ \hline 1 & 0 \\ \hline 0 & 1 \\ \hline \end{array} \\ & A & \begin{array}{|c|c|c|} \hline 3 & 3 & -1 \\ \hline 2 & 1 & 3 \\ \hline \end{array} & \begin{array}{|c|c|} \hline & 2 \\ \hline 3 & 5 \\ \hline \end{array} \\ & & & A \cdot B \end{matrix}$$

Matrix Multiplication

Exercise 15

A

3	3	-1
2	1	3

B

1	1
1	0
0	1

5	4	2
3	3	-1
2		3

$B \cdot A$

Matrix Multiplication

Exercise 16

$$\begin{matrix} & & & B \\ & & & \begin{array}{|c|c|} \hline 3 & 2 \\ \hline 2 & 1 \\ \hline 1 & -2 \\ \hline \end{array} \\ & A & \begin{array}{|c|c|c|} \hline 0 & 1 & \\ \hline 1 & 1 & 0 \\ \hline \end{array} & \begin{array}{|c|c|} \hline 3 & -1 \\ \hline 5 & 3 \\ \hline \end{array} \\ & & & A \cdot B \end{matrix}$$

Matrix Multiplication

Exercise 17

$$\begin{matrix} & & & & & B \\ & & & & \begin{array}{|c|c|} \hline -2 & \\ \hline 4 & 4 \\ \hline 3 & 1 \\ \hline \end{array} \\ & & & & \\ A & \begin{array}{|c|c|c|} \hline 1 & 0 & 1 \\ \hline 1 & 1 & 0 \\ \hline \end{array} & \begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & 6 \\ \hline \end{array} \\ & & A \cdot B \end{matrix}$$

Matrix Multiplication

Exercise 18

$$\begin{matrix} & & & & B \\ & & & & \begin{array}{|c|c|} \hline 3 & 2 \\ \hline 2 & 0 \\ \hline 1 & 2 \\ \hline 4 & 3 \\ \hline \end{array} \\ & A & \begin{array}{|c|c|c|c|} \hline 1 & 1 & 0 & 1 \\ \hline 1 & 0 & 1 & -1 \\ \hline \end{array} & \begin{array}{|c|c|} \hline 9 & \text{ } \\ \hline 0 & 1 \\ \hline \end{array} \\ & & & A \cdot B \end{matrix}$$

Matrix Multiplication

Exercise 19

A

1	1	0	1
1	0	1	-1

B

3	2
2	0
1	2
4	3

5	3	2	1
2	2	0	2
3	1		-1
7	4	3	1

B · A

Matrix Multiplication

Exercise 20

				B		
				3	2	
				2	1	
				1	1	
				4	3	
A	1	1	0	0	5	3
	0	1	1	0	3	
	0	0	1	1	5	4
	1	0	0	1	7	5
A·B						

Matrix Multiplication

Exercise 21

B

3	2
2	1
1	1
4	3

A

1	-1	0	0	1	1
0	1	-1	0	1	0
0	0	1	-1		-2
-1	0	0	1	1	1

$A \cdot B$

-3

Matrix Multiplication

Exercise 22

B

1	0	0	0	1
0	1	0	0	1
0	0	1	0	0
0	0	0	1	0

A

9	-2	0	5
13	3	1	4

9		0	5	7
13	3	1	4	16

A·B

-2

Matrix Multiplication

Exercise 23

B

0	0	0	1	1
1	0	1	0	1
0	1	1	0	1
0		0	1	0

A

2	-1	1	0	-1	1	0	2	0
3	2	4	1	2	5	6	4	9

$A \cdot B$

Matrix Multiplication

Exercise 24

B

3	2
2	1
1	1
4	3

A

-1	1	0	0	-1	-1
0	1	0	0	2	1
	0	1	0	-2	-1
1	0	0	1	7	5

$A \cdot B$

-1

Matrix Multiplication

Exercise 25

B

0	0	0	1	1
1	0	1	0	1
0	1	1	0	1
0	0	0	1	0

A

1	1	1	0
0	1	1	1
1	0	1	1
1	1	0	1
1	1	-1	1

1	1	2	1	3
1	1		1	2
0	1	1	2	2
1	0	1	2	2
1	-1	0	2	1

$A \cdot B$